**Practical 8**

**Enroll No : 2508433116002**

**Aim : Create interactive visualizations using Plotly Express and Plotly Graph Objects.**

**step - 1 import the requred libraries**

```
#import pandas library
import pandas as pd
#import plotly express library
import plotly.express as px
#import plotly graph library
import plotly.graph_objects as go
```

**step -2 create the dataset**

```
#create dataset
data = {
    "Month": ["januray", "february", "march", "april", "may", "june", "july", "august", "september", "october", "novembe
    "Product": ["Laptop", "Tablet", "mobile", "Laptop", "Tablet", "mobile", "Laptop", "Tablet", "mobile", "Laptop", "Tab
    "Region": ["north", "south", "east", "west","north", "south", "east", "west","north", "south", "east", "west"],
    "sales": [120000,135000,150000,145000,160000,175000,180000,170000,190000,210000,230000,250000],
    "Units_Sold": [12, 30, 25, 14, 35, 30, 18, 38, 32, 21, 45, 40],
    "Profit": [18000, 27000, 30000, 22000, 32000, 35000, 28000, 34000, 38000, 33000, 46000, 50000]
}
df = pd.DataFrame(data)
df
```

| | Month | Product | Region | sales | Units_Sold | Profit |
|---|---|---|---|---|---|---|
| 0 | januray | Laptop | north | 120000 | 12 | 18000 |
| 1 | february | Tablet | south | 135000 | 30 | 27000 |
| 2 | march | mobile | east | 150000 | 25 | 30000 |
| 3 | april | Laptop | west | 145000 | 14 | 22000 |
| 4 | may | Tablet | north | 160000 | 35 | 32000 |
| 5 | june | mobile | south | 175000 | 30 | 35000 |
| 6 | july | Laptop | east | 180000 | 18 | 28000 |
| 7 | august | Tablet | west | 170000 | 38 | 34000 |
| 8 | september | mobile | north | 190000 | 32 | 38000 |
| 9 | october | Laptop | south | 210000 | 21 | 33000 |
| 10 | november | Tablet | east | 230000 | 45 | 46000 |
| 11 | december | mobile | west | 250000 | 40 | 50000 |

Next steps: Generate code with df | New interactive sheet

**Step 3: Inspect the Dataset**

```
#display first five records from the dataset
df.head()
```

| | Month | Product | Region | sales | Units_Sold | Profit |
|---|---|---|---|---|---|---|
| 0 | januray | Laptop | north | 120000 | 12 | 18000 |
| 1 | february | Tablet | south | 135000 | 30 | 27000 |
| 2 | march | mobile | east | 150000 | 25 | 30000 |
| 3 | april | Laptop | west | 145000 | 14 | 22000 |
| 4 | may | Tablet | north | 160000 | 35 | 32000 |

Next steps: Generate code with df | New interactive sheet

```
#display information about the dataset
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12 entries, 0 to 11
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype 
---  ------      --------------  ----- 
 0   Month       12 non-null     object
 1   Product     12 non-null     object
 2   Region      12 non-null     object
 3   sales       12 non-null     int64 
 4   Units_Sold  12 non-null     int64 
 5   Profit      12 non-null     int64 
dtypes: int64(3), object(3)
memory usage: 708.0+ bytes
```

```
#display rows and columns
df.shape
```

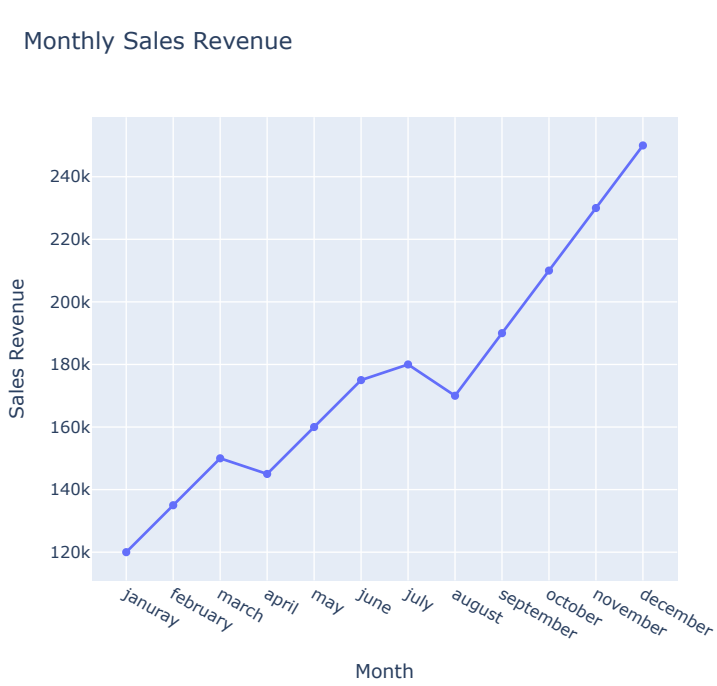
```
(12, 6)
```

**Step - 4 create an interective line chart using plotly express**

```
fig = px.line(
    df,
    x='Month',
    y='sales',
    markers=True,
    title='Monthly Sales Revenue'
)

fig.update_layout(
    xaxis_title='Month',
    yaxis_title='Sales Revenue',
    hovermode='x unified'
)

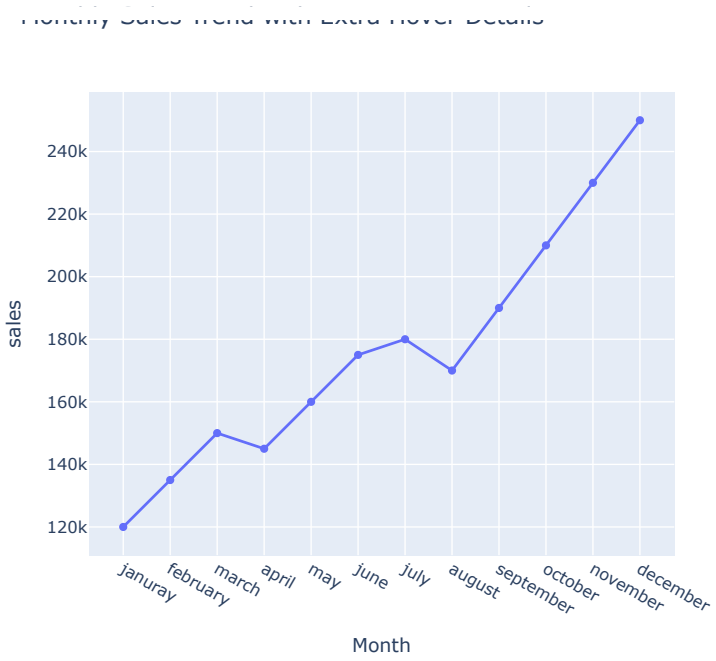
fig.show()
```




**step - 5 improve the hover information**

```
import plotly.express as px
fig = px.line(df, x="Month", y="sales", markers=True, title="Monthly Sales Trend with Extra Hover Details", hover_data=[
```

```
fig.show()
```




**Step 6: Create an Interective Bar Chart**

```
# Group the dataset by Product and calculate total Sales
product_sales = df.groupby("Product", as_index=False)["sales"].sum()

# Create interactive bar chart
fig = px.bar(
    product_sales,
    x="Product",
    y="sales",
    title="Total Sales Generated by Each Product",
    text="sales",
    hover_data=["sales"]
)

fig.update_layout(
    xaxis_title="Product",
    yaxis_title="Total Sales",
    hovermode="x"
)

fig.show()
```

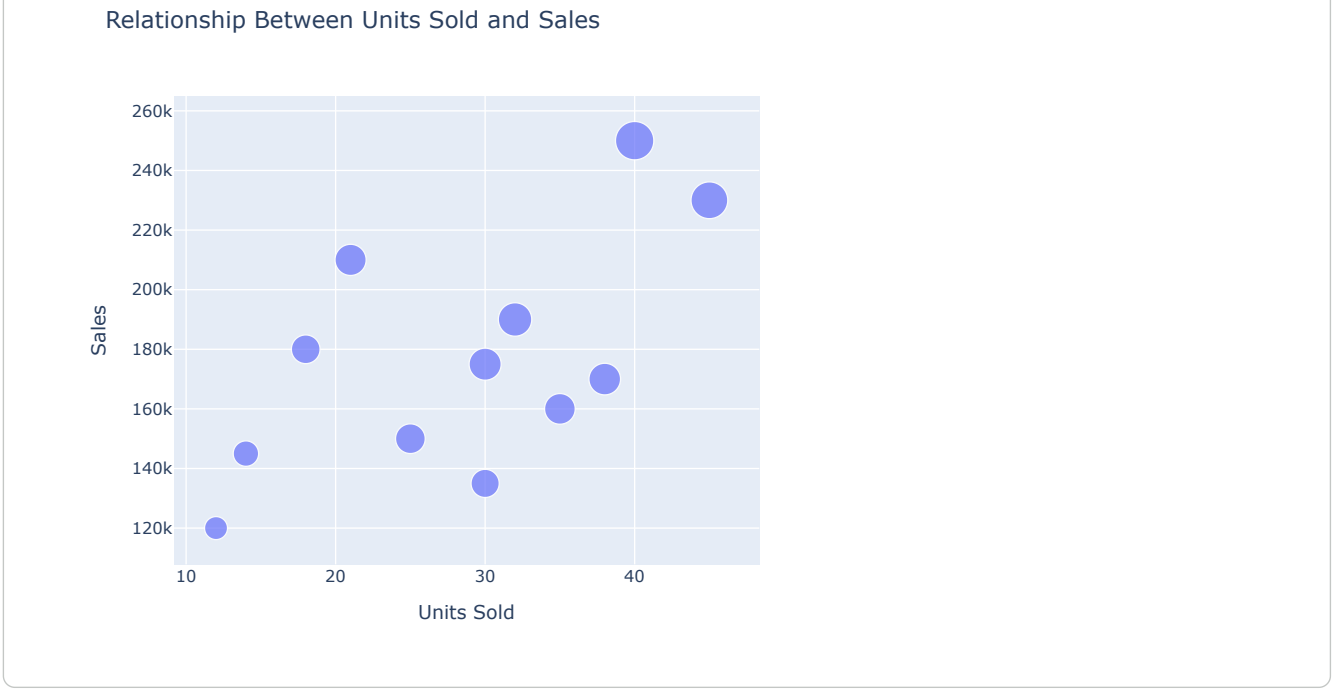



**Step 7: Interective Scatter Plot**

```
fig = px.scatter(
    df,
    x="Units_Sold",
    y="sales",
    title="Relationship Between Units Sold and Sales",
    hover_data=["Month", "Product", "Region", "Profit"],
    size="Profit"
)

fig.update_layout(
    xaxis_title="Units Sold",
    yaxis_title="Sales",
    hovermode="closest"
)

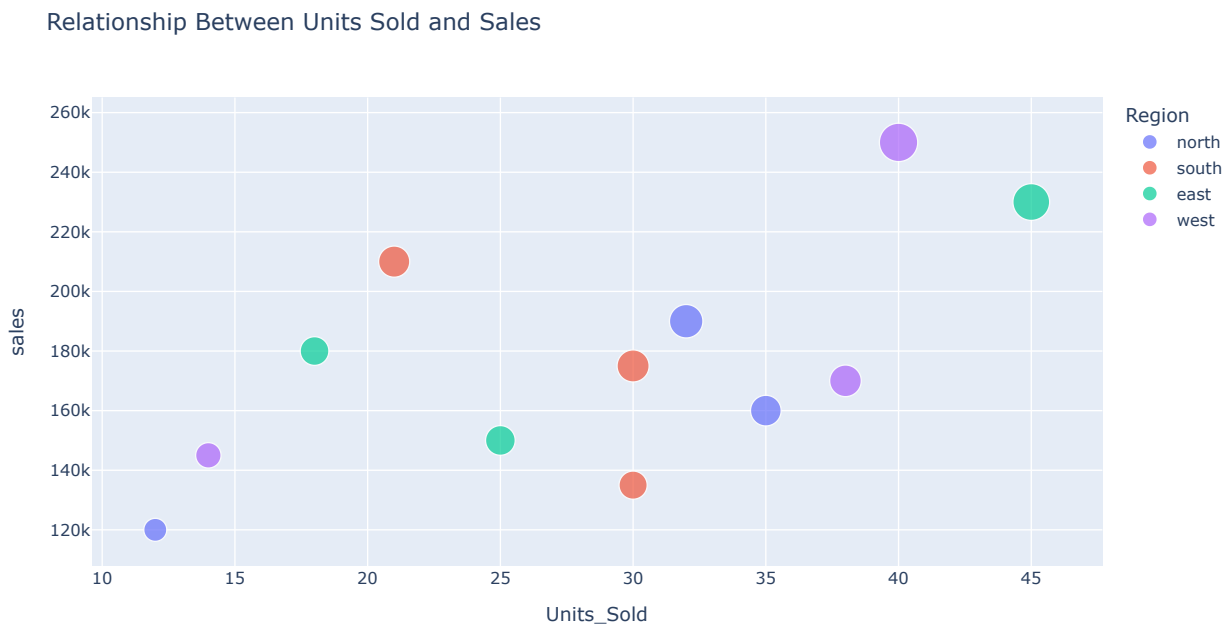
fig.show()
```



**step 8 interactive scatter plot with color**

```
#create interactive scatter plot of unit_sold and sales with colors.
fig = px.scatter(
    df,
    x="Units_Sold",
    y="sales",
    color="Region",
    title="Relationship Between Units Sold and Sales",
    hover_data=["Month", "Product", "Region", "Profit"],
    size="Profit",

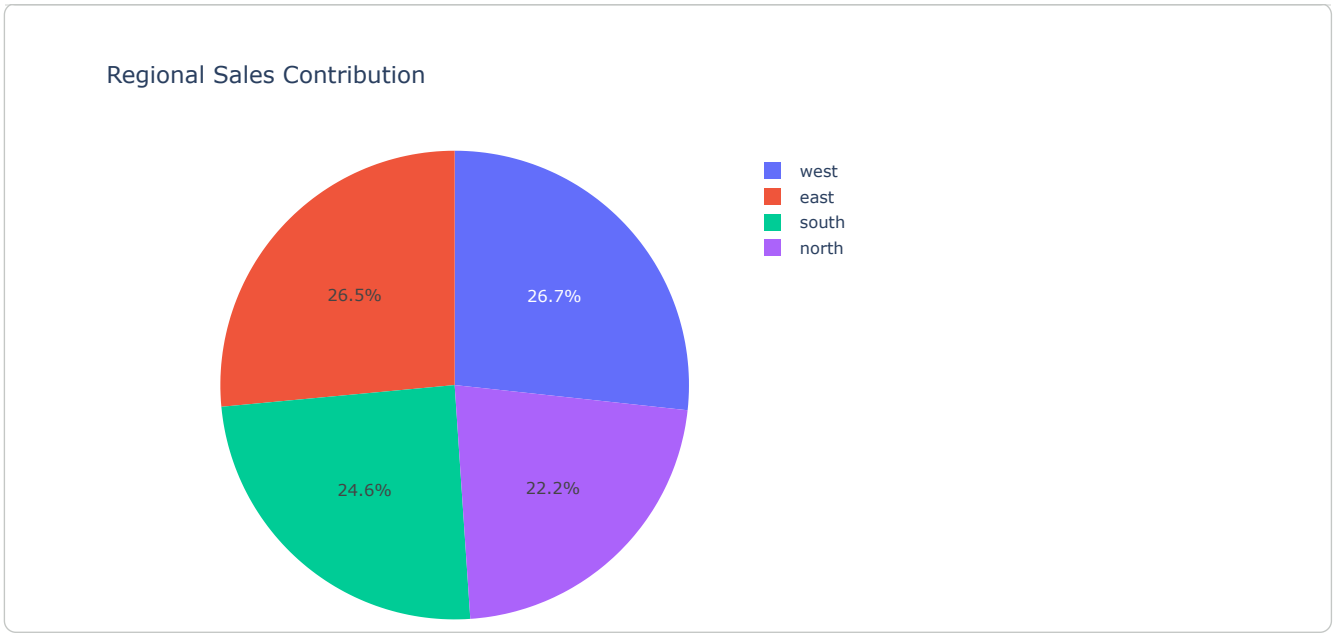
    )
fig.show()
```




**step 9 create an interective pie chart**

```
# Calculate total sales of region
region_sales = df.groupby("Region", as_index=False)["sales"].sum()

# Create pie chart
fig = px.pie(
    region_sales,
    names="Region",
    values="sales",
    title="Regional Sales Contribution",
    hover_data=["sales"],
    hole=0
)
fig.show()
```




**Step - 10 save the dataset**

```
# Save dataset
df.to_csv("DAta_Analysis.csv", index=False)

print("CSV file saved Successfully")
# Download CSV file.
from google.colab import files

files.download("DAta_Analysis.csv")
```

```
CSV file saved Successfully
```